Final Project – Introduction to Data Science

Submitted by: Shaked Maman

1. Introduction

Because of my great love for music, I looked for a dataset that contains information about music in one form or another. I find a lot of interest in creating music and building playlists, and for years I have been playing classical guitar and making electronic music. I once thought to myself that if I could find a way to make a living from creating playlists, I would make a good living. When I discovered the world of data analysis in music, I realized that a bachelor's degree in statistics and data analysis could be an interesting path to working behind the scenes of the music industry, especially in the world of streaming. Through TidyTuesday I found a dataset of songs from the Spotify database, which contains information about a large number of songs. This dataset includes interesting fields, such as a score for how acoustic a song is, a "danceability" score (how suitable a song is for dancing), an energy score, and simpler fields such as genre, album name, artist name and so on. This dataset makes it possible to ask interesting research questions and to look for connections between different characteristics.

I really enjoyed working on this project, and I am happy that I managed to find a dataset that I find interesting. I hope you will enjoy reading it as well.

In the introduction I also added a table that explains the meaning of each field in the dataset and how the data is presented: how each score is calculated, what it represents, what its range of values is, and what a high or low value means. The goal is to help the reader understand what is being measured and what the final results mean.

Throughout the project I refer to a term called the "Streaming Era". Because of this, I felt the need to write a short paragraph about this period, explaining which years it covers and what it actually means:

The Streaming Era refers to the period in which music distribution moved mainly to online platforms that allow direct listening (streaming), without the need to download or physically buy albums. It began in the late 2000s, with the appearance of services such as Spotify (launched in 2008) and Apple Music (2015), and it is characterized by almost immediate access to a huge catalog of songs, a subscription-based business model, and the ability of independent artists to reach a wide audience without going through traditional record labels. This period continues to this day and represents a fundamental change in listening habits, in song length and in the pace of music production.

Preparing the work environment:

1. The Spotify data table, taken from TidyTuesday: https://github.com/rfordatascience/tidytuesday/tree/main/data/2020/2020-01-21
2. The original table contains a column called duration_ms, which holds the length of each song in milliseconds. I wanted to convert this column to seconds, so I also changed the column name from duration_ms to duration_sec.

The next page shows a table describing each feature ("field") in the original dataset. The table explains what each feature means and how it is measured.

spotify_songs.csv

variable	class	description
track_id	character	Song unique ID
track_name	character	Song Name
track_artist	character	Song Artist
track_popularity	double	Song Popularity (0-100) where higher is better
track_album_id	character	Album unique ID
track_album_name	character	Song album name
track_album_release_date	character	Date when album released
playlist_name	character	Name of playlist
playlist_id	character	Playlist ID
playlist_genre	character	Playlist genre
playlist_subgenre	character	Playlist subgenre
danceability	double	Danceability describes how suitable a track is for dancing based on a combination of musical elements including tempo, rhythm stability, beat strength, and overall regularity. A value of 0.0 is least danceable and 1.0 is most danceable.
energy	double	Energy is a measure from 0.0 to 1.0 and represents a perceptual measure of intensity and activity. Typically, energetic tracks feel fast, loud, and noisy. For example, death metal has high energy, while a Bach prelude scores low on the scale. Perceptual features contributing to this attribute include dynamic range, perceived loudness, timbre, onset rate, and general entropy.
key	double	The estimated overall key of the track. Integers map to pitches using standard Pitch Class notation . E.g. 0 = C, 1 = C♯/D♭, 2 = D, and so on. If no key was detected, the value is -1.
loudness	double	The overall loudness of a track in decibels (dB). Loudness values are averaged across the entire track and are useful for comparing relative loudness of tracks. Loudness is the quality of a sound that is the primary psychological correlate of physical strength (amplitude). Values typical range between -60 and 0 db.
mode	double	Mode indicates the modality (major or minor) of a track, the type of scale from which its melodic content is derived. Major is represented by 1 and minor is 0.
speechiness	double	Speechiness detects the presence of spoken words in a track. The more exclusively speech-like the recording (e.g. talk show, audio book, poetry), the closer to 1.0 the attribute value. Values above 0.66 describe tracks that are probably made entirely of spoken words. Values between 0.33 and 0.66 describe tracks that may contain both music and speech, either in sections or layered, including such cases as rap music. Values below 0.33 most likely represent music and other non-speech-like tracks.
acousticness	double	A confidence measure from 0.0 to 1.0 of whether the track is acoustic. 1.0 represents high confidence the track is acoustic.
instrumentalness	double	Predicts whether a track contains no vocals. "Ooh" and "aah" sounds are treated as instrumental in this context. Rap or spoken word tracks are clearly "vocal". The closer the instrumentalness value is to 1.0, the greater likelihood the track contains no vocal content. Values above 0.5 are intended to represent instrumental tracks, but confidence is higher as the value approaches 1.0.
liveness	double	Detects the presence of an audience in the recording. Higher liveness values represent an increased probability that the track was performed live. A value above 0.8 provides strong likelihood that the track is live.
valence	double	A measure from 0.0 to 1.0 describing the musical positiveness conveyed by a track. Tracks with high valence sound more positive (e.g. happy, cheerful, euphoric), while tracks with low valence sound more negative (e.g. sad, depressed, angry).
tempo	double	The overall estimated tempo of a track in beats per minute (BPM). In musical terminology, tempo is the speed or pace of a given piece and derives directly from the average beat duration.
duration_ms	double	Duration of song in milliseconds

2. Methods and Results

2.1 - Is there a relationship between a song's tendency to make people dance and its popularity?

To answer this research question, I used a TidyTuesday dataset of songs played on the Spotify platform. The table I downloaded includes columns with scores for danceability and popularity. The relevant fields are:

danceability – a score between 0 and 1, where a high value is given to a song that tends to make people dance.

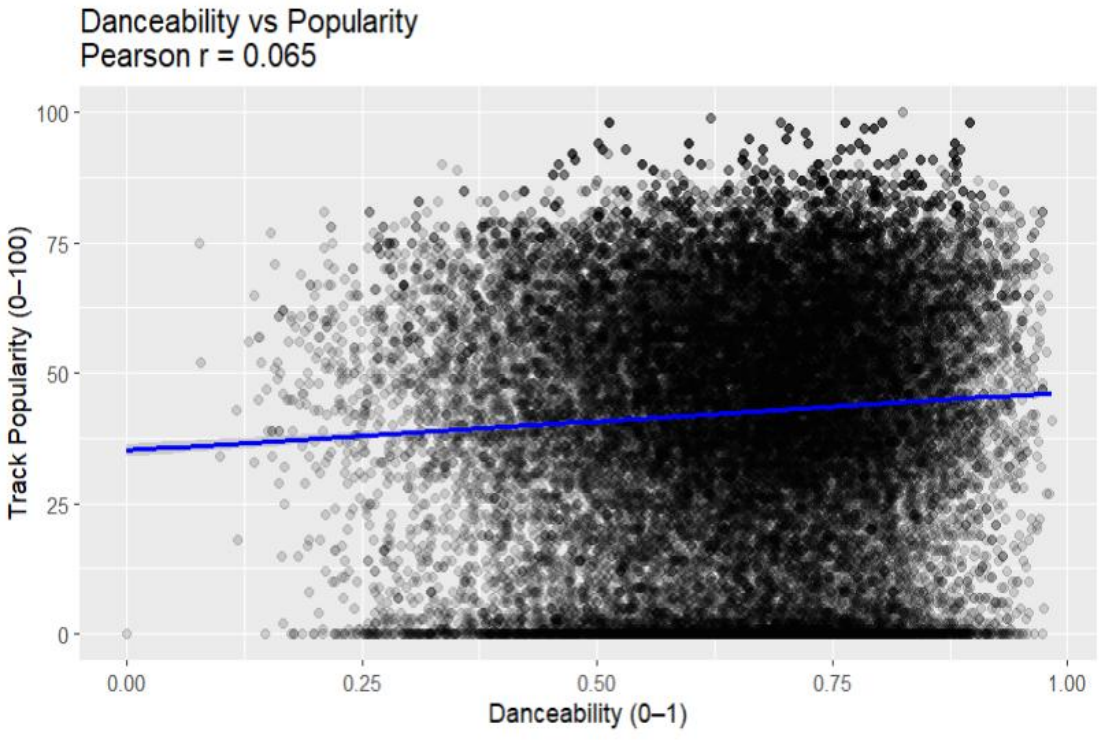
track_popularity – a popularity score between 0 and 100, where a high value represents higher popularity.

Therefore, the estimand in this question is the relationship between a song's tendency to make people dance and its popularity. In my opinion, the most suitable estimator for answering this research question is the Pearson correlation coefficient, since it is a statistical tool that measures the strength and direction of a linear relationship between two quantitative variables. This way we can understand whether, when one variable increases, the other one also tends to increase (a positive relationship) or to decrease (a negative relationship).

My hypothesis for this research question is that there is a positive relationship between a song's tendency to make people dance and its popularity. This assumption comes from my feeling that a "danceable" song makes the listener happy, so the listener will tend to give the song a high rating. As a result, its popularity will increase.

I will present the findings in a scatter plot, where the X axis represents the song's danceability (a score between 0 and 1), and the Y axis represents the song's popularity (a score between 0 and 100).

Findings:



The correlation value obtained is r = 0.065. This value represents a positive but very weak correlation. It does show a small general tendency for songs with higher danceability to have higher popularity, but this relationship is almost negligible from a statistical point of view.

The points are spread fairly evenly along the danceability axis, which shows that there is no sharp concentration of values only at low or only at high danceability. Along the Track Popularity axis, it is hard to point to any special concentration in specific areas of the graph.

The blue trend line shows a positive slope, but a very gentle one. In other words, as danceability increases, the song's popularity increases slightly, but the slope is very gentle.

Main conclusion:

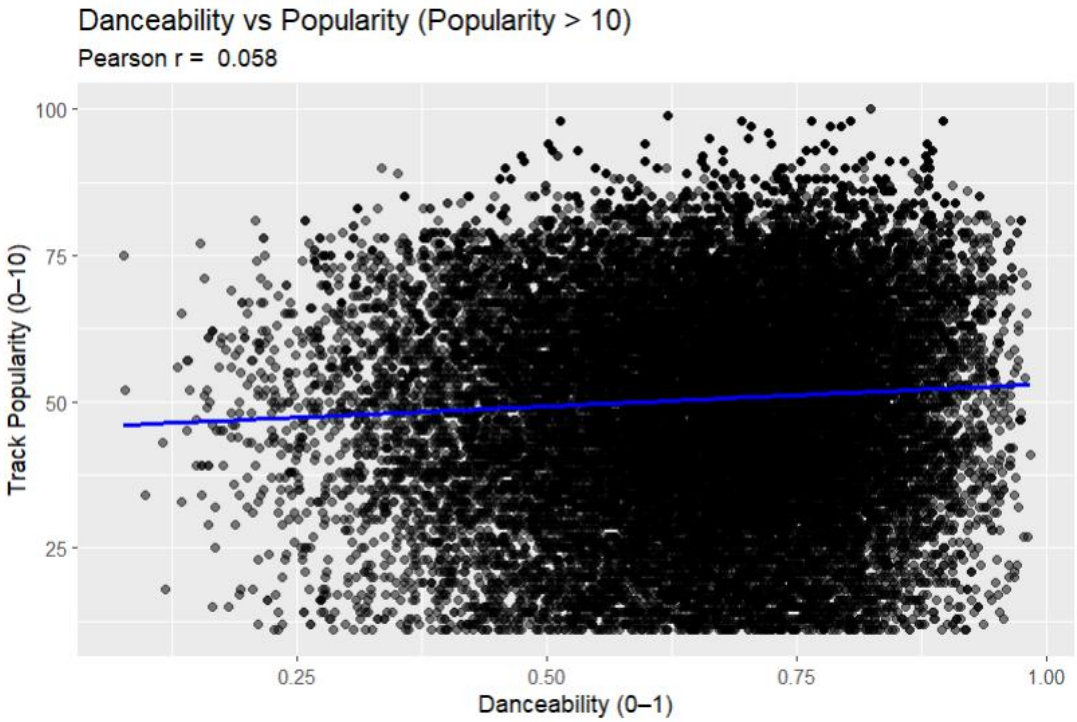
Danceability alone is a very weak indicator of a song's popularity in this dataset. From this we can conclude that the danceability score does not indicate a song's popularity, and that there are probably other factors that affect popularity, such as marketing, the artist's name, the release year and so on. Visually, the data shows a wide spread with no clear pattern, so in practical terms it is difficult, if not impossible, to rely on danceability alone to predict popularity.

Since there was a large concentration of values with popularity = 0, I was interested in checking the research question again after filtering out songs with minimal popularity. To take the research question further, I will filter out songs with popularity lower than 10 and check whether the relationship exists.

My hypothesis here is that the correlation coefficient will increase in the positive direction, and that we will be able to point to a clearer positive linear relationship.

I checked the dataset and found that there are 5,243 songs with a popularity value of 0, which is almost one sixth of the dataset. Therefore, this seems like an interesting direction to explore.

Findings:



Conclusions:

After filtering the songs so that only songs with popularity above 10 remained, the following correlation was obtained: r = 0.058. Not only is this correlation very weak, it is even weaker than in the original analysis (r = 0.065). The scatter plot shows a wide spread of danceability values across the entire range of the variable, and the linear trend line shows an extremely gentle positive slope. My hypothesis was that this step would increase the Pearson correlation coefficient, assuming that songs with low or zero popularity act as noise in the research question. In practice, not only was there no improvement, the correlation coefficient actually decreased. This strongly supports the conclusion of the research question, and we can say that the danceability score alone is not a meaningful predictor of popularity, even among songs with relatively high popularity.

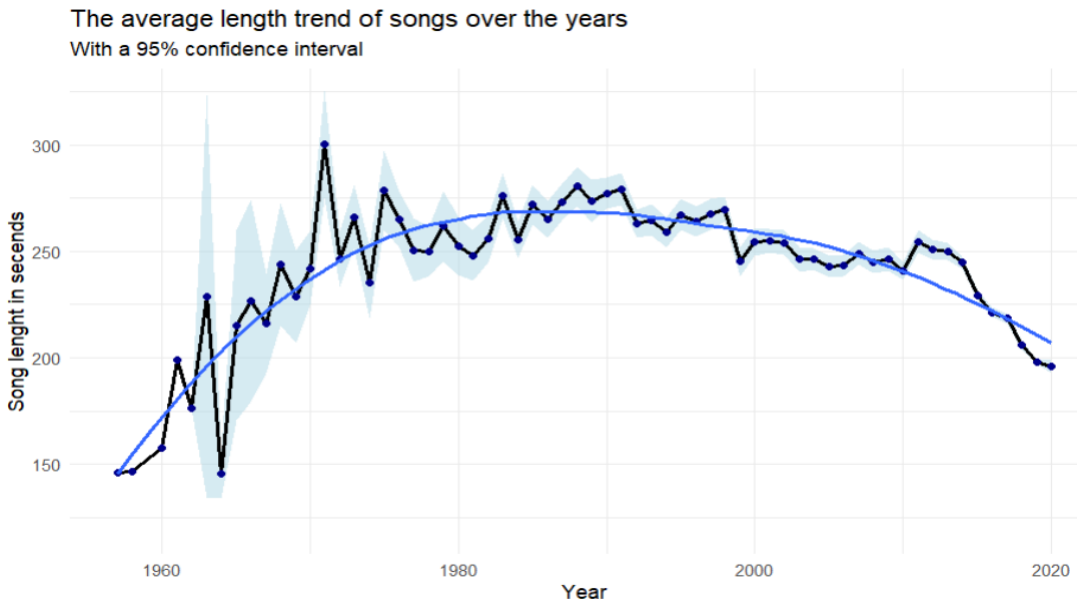
2.2 - Has the average length of songs (in seconds) changed over the years, and if so, in what way?

To answer this question, I used a package that was not covered in the practice sessions: the stringr package. This package helps me create a column with the release year of each song. The dataset has a column with release dates in slightly different formats, but all the dates start with the release year. The **str_sub** function lets me take text from a chosen column and return only the first 4 characters, which are the release year. The release year is a string object, so I use as.numeric() to convert it into a numeric value.

My hypothesis for this research question is that the average length of songs released in different years will change, in my opinion according to what listeners wanted in each period. For example, I know that in the era when rock music developed, songs called "rock ballads" were released, and they were over seven minutes long (today almost no songs come close to this length, and it is even common to release songs that are half as long). For example, "Shine On You Crazy Diamond" by Pink Floyd is about 13 minutes long, and the whole album it appears on, "Wish You Were Here", has songs of similar lengths. Therefore, my hypothesis is that there will be significant changes in song length between periods, and this is my motivation for this research question.

In this question, my estimand is the average length of a song in seconds in each year, and my estimator (which is actually the method) is the average length of the songs in the sample for each year.

Findings:



Conclusions:

The analysis shows a clear trend in the average song length over the years. A pattern with three fairly clear stages was observed:

In the 1950s the average length was only about 150 seconds, and over the next two decades there was a **sharp and continuous rise**, up to a peak of about 300 seconds in the 1970s. From the 1980s to the early 2000s there was **relative stability**, with an average of around 250–270 seconds. From the mid-2000s there is a **gradual decline**, which in recent years became a sharp decline, down to only about 200 seconds around 2020. The smooth trend line (loess) confirms this pattern and highlights three main stages: rise, stability and decline.

The 95% confidence band makes it possible to assess whether the differences between years are significant. When the bands of different years do not overlap, we can conclude that the differences between them are statistically significant. In this case, the difference between the 1950s and the 1970s, as well as between the late 2000s and 2010–2020, shows almost no overlap, which supports the hypothesis of a significant change. In contrast, during the stable period (the 1980s to the early 2000s) the confidence bands overlap a lot, which suggests that there was no meaningful change during this period.

The findings support the hypothesis that song length changed significantly over the years, and suggest that cultural, technological and commercial changes may explain the gradual lengthening in the second half of the 20th century and the noticeable shortening in the Streaming Era. It is interesting to look at the trends by period, and to think about what listeners were interested in at the time and how music producers created their music accordingly.

2.3 - Is there a significant difference in popularity between genres over the years, and does the major/minor mode of a genre have an effect?

For this research question I used the Spotify songs dataset that I took from TidyTuesday.

In this research question I will check, based on this dataset, which genre is the most popular. To make the question a bit more interesting, I decided to also check whether the "mode" of the songs has an effect, meaning whether the genre is more major or more minor. The "mode" of a song depends on the type of scale it uses, and songs can be roughly divided into two types (major/minor). A major song usually sounds happier and more cheerful, while a minor song usually sounds quieter.

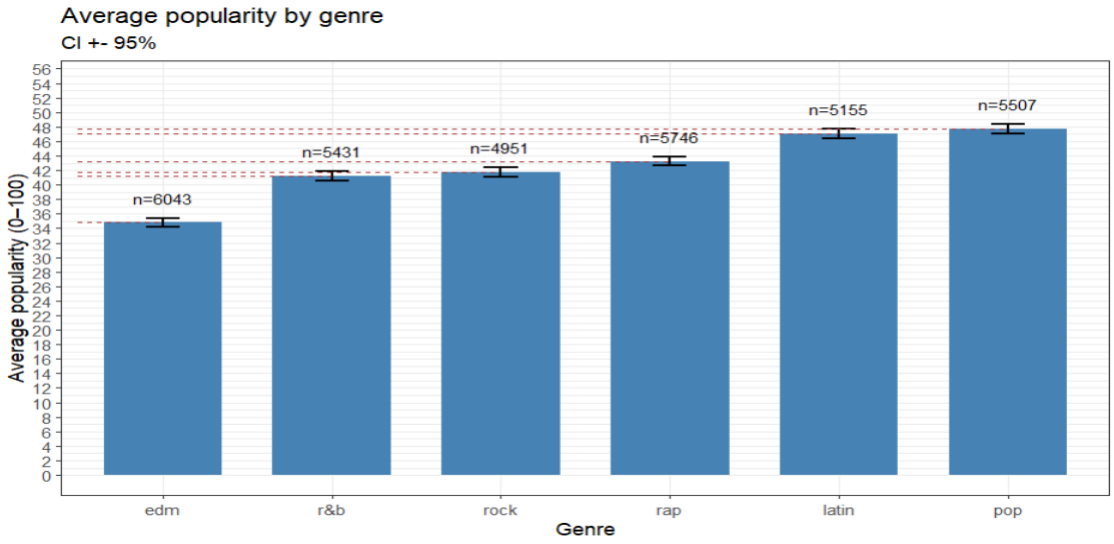
I decided in advance that for a genre to be called major or minor, at least 80% of the songs in that genre must be major, or minor. I will measure this using a field in the table called **mode**. The field is binary, where the value 1 represents a major song and the value 0 represents a minor song. If the condition is not met, the genre will be defined as "mixed".

Method: First I will check the popularity of the genres, and then the effect of the genre's "mode", if a genre can be labeled as major or minor at all. This is my estimand. Using the average popularity of all songs in each genre, I will check which genre is the most popular. This is my estimator for this research question. The goal is to see which genres really dominate the industry, and whether the difference between the leading genres is significant. It is interesting to see whether there is a dominant genre, or whether the difference between the average popularity values is negligible. After that, I will check the "mode" of the genres. If at least 80% of a genre's songs are in a certain "mode", I can say that the genre is major or minor, and determine, beyond which genres are more popular, whether major/minor has an effect.

My research hypothesis is that the POP genre will be the most popular. I feel that over the years many artists aimed for this style, and that it works for private listening, social settings and even clubs. I think ROCK will come next, because it is a genre that covers a wide range of years, so I feel that a lot of people have connected to it over a very long period of time.

I will present the data in a bar chart. First, I will show the average popularity by genre:

Findings:



In the graph we can see the average popularity of each genre. As mentioned, a song's popularity score is between 0 and 100, where 100 is the highest rating and 0 is the lowest. The dashed red line marks the average itself, and I added it to make the average easier for the reader to see. In

addition, for each bar (genre), I added the number of observations it has in the dataset (the number of songs).

The graph shows clear descriptive differences in average popularity between genres. The pop and latin genres show high averages (between 47–49), while edm shows a lower average (35). The error bars are narrow because of the large sample sizes (4–6 thousand songs per genre), and between most genres there is almost no overlap, which suggests that the ranking is stable. In contrast, between r&b and rock there is a wider overlap, so the difference between them is small. Overall, the gap between the highest and lowest genre is about 13–14 points, which seems meaningful descriptively, although differences of 1–2 points may be negligible in practice.

The results partly confirm my hypothesis, since POP is indeed the leading genre. Right after it comes the LATIN genre. In my opinion, this is because the genre is popular all over the world, and also because Latin populations listen to a lot of music, especially Latin music. I felt this during my big trip in South and Central America.

Now I will present a table showing the average "majorness" of each genre, meaning how major each genre actually is (the data appears in the "mode_major_mean" column, where the value is shown as a percentage):

	playlist_genre	mean_pop	sd_pop	mode_major_mean
1	edm	34.83353	23.15423	52.01059
2	latin	47.02658	25.42456	56.19787
3	pop	47.74487	25.15833	58.81605
4	r&b	41.22353	25.89450	52.14509
5	rap	43.21545	23.30209	52.14062
6	rock	41.72834	24.82523	70.02626

According to the condition I defined at the beginning of the research question, no genre can be clearly called major or minor. Based on this condition, all the genres are defined as mixed, and overall no genre seems to have a clear tendency toward major. Therefore, this feature cannot be considered a factor that affects genre popularity.

2.4 - Has the average number of songs released each year changed significantly between different decades?

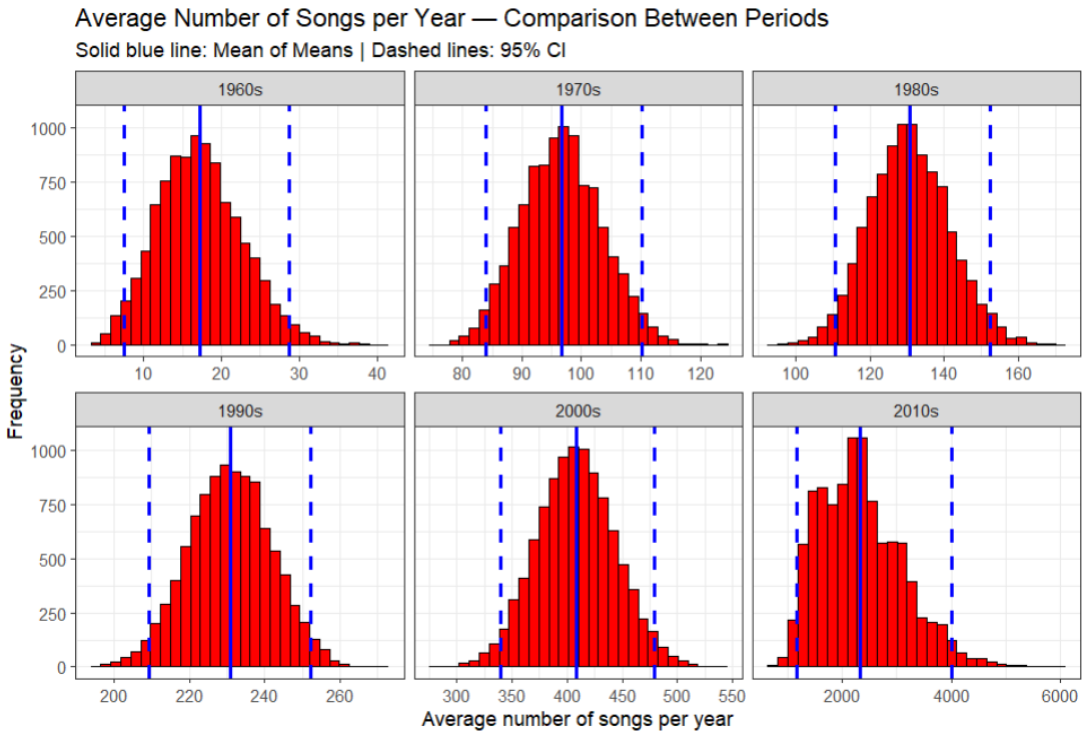
In the Streaming Era, platforms like Spotify allow large-scale analysis of musical trends over time. One of the central questions about how this data can be used is the number of songs released each year, and whether there is a consistent pattern or unusual fluctuations over the years. Examining the average number of songs per year helps us understand the pace of music production and the level of variability between different years. However, the mean calculated from the sample may be affected by the sample size and the shape of the distribution. To deal with this uncertainty, I used a resampling method (Bootstrap), which makes it possible to estimate the empirical distribution of the mean and build confidence intervals around it. This way, we can estimate not only the mean itself, but also how stable and reliable it is.

The goal of this research question is to examine the pace of music production over the years by calculating the average number of songs per year. To present the results in a way that is easier to analyze and understand, I will divide the years into time periods (periods) by decade, so that all songs released between 1960 and 1970 will be called **1960s**. My estimator is the sample mean of

the number of songs per year in each decade. My estimand is the number of songs released per year in each decade.

Through this analysis we can compare the average number of songs per year, group the years into periods, and see whether the number of songs released in each period is relatively stable over the years, or whether there is noticeable variability between periods.

I will present the results in graphs, where each graph shows the distribution of the average number of songs per year in that decade, calculated using bootstrap. The number of repetitions (resamples) I used for each sample is 10,000. **Results:**



I presented the results using histograms. On each histogram I added the estimated mean for that decade (the mean of the means), shown as a solid blue line. In addition, on each histogram I added the bounds of the 95% confidence interval, shown as two dashed blue lines.

Analyzing the average number of songs per year by decade shows a clear pattern of continuous growth over the periods. In the 1960s, the average number of songs per year was only about 20, with a narrow confidence interval that indicated a stable estimate. In the 1970s there was a sharp increase to an average of about 100 songs per year, and in the 1980s the average rose to about 140 songs per year. In the 1990s there was another jump, with an average of around 230 songs per year. The growth became stronger in the 2000s, with an average of about 400–450 songs per year, and in the 2010s there was an exceptional acceleration in the amount of music created, with an average of about 2,000 songs per year.

The confidence intervals obtained using Bootstrap were relatively narrow in all periods, which indicates that the differences between the decades are significant and are not just a result of sampling variability. The findings clearly point to an accelerating growth trend in the number of songs produced on digital platforms, a trend that reached its peak in the last decade.

Below is a table that summarizes the following information for each time period:

	period	mean_of_means	ci_lo	ci_hi
1	1960s	17.23292	7.50	28.7000
2	1970s	96.65285	83.90	110.2000
3	1980s	130.64965	110.50	152.3000
4	1990s	230.77933	209.30	252.1000
5	2000s	408.13005	340.10	478.8025
6	2010s	2333.77394	1173.78	4016.3975

Discussion

The findings point to an accelerating increase in the number of songs per year over the decades, a trend that peaks in the 2010s. This result can be explained in the broader context of how the music industry developed: the shift from physical production to digital distribution, the rise of streaming platforms such as Spotify, and the ability of independent artists to release their work on their own at a low cost. In addition, the growing demand for music on online platforms created an incentive for a dramatic increase in new releases.

Limitations of the research question

It is important to note that the dataset used is based on Spotify data, so it may not fully reflect all the songs released each year, especially in earlier decades (such as the 1960s and 1970s). The platform has a richer representation of songs from later periods, while songs from earlier periods may never have been documented or uploaded to Spotify. This may create an observation bias, so the low values found in these years do not necessarily reflect low music production, but rather missing information in the dataset. Because of this limitation, the focus should mainly be on identifying the relative trends over time rather than on the absolute values.

3. Discussion

This data analysis shows how the Spotify dataset allowed me to turn my personal curiosity about the world of music into a broad empirical analysis. The findings show that the danceability score alone hardly predicts popularity, which highlights that listeners' connection to songs comes from a complex combination of cultural, marketing and artistic factors. Regarding song length, I found a clear pattern of growth in the second half of the 20th century, then relative stability, and then a sharp decline in the Streaming Era – a trend that reflects changes in listening habits and in the nature of music production. When comparing genres, POP was indeed found to lead in popularity, but no significant effect of major or minor mode was found. Finally, examining the number of songs released each year showed dramatic growth, mainly from the 2000s onward, which reflects the shift to digital distribution and the growing accessibility of the creative process.

These results not only support the hypotheses I set, but also show how a personal love for music can be combined with statistical tools and data analysis skills. For me, this project is a concrete example of how the world of statistics can enrich our understanding of the music industry and reveal the patterns behind the musical experiences we all share.